

VBF-T1R1NOFORCE-DVPR-01 - Design Verification Plan and Report (virtual-test version)

VBF-T1R1NOFORCE-DVPR-01 · Generation date: 2026-08-25 · All results are simulation outputs (PyBaMM cell model + thermal-runaway ODE)

#	Verification item	Condition	Result value	Determination (vs criteria)
1	1C discharge capacity	1C CC, 25 °C ambient, lumped thermal, to 2.5 V	7.654 Ah	Informational (no task)
2	Energy density	contract formula: $SV \cdot I \cdot dt \div \text{SUM thickness} \times (1 - 0.05) \times 2 \text{ Wh/kg area}$	457.93 Wh/kg	[PASS] PASS (≥ 392.61)
3	4C fast-charge temperature	4C CC at 45 °C ambient, lumped thermal, with plating model	318.79 K	[PASS] PASS (≤ 333.15)
4	4C fast-charge lithium plating	lithium plating potential vs Li/Li+, plating if any value $< 0.00824 \text{ V (SPMe)}; +0.00515 \text{ V (DFN cross-check)}$	never	[PASS] PASS (never)
5	Overcharge -> thermal runaway	1C CC charge to 4.7 V (4.2 + 0.5), then three-side triggered ODE (mass=4.728g, 22 K, dT/dmass=900)	no thermal runaway	[PASS] PASS (no thermal runaway)
6	Voltage window parameter set		2.5 – 4.2 V (overcharge window verified to 4.7 V)	Informational
7	4C CC charge acceptance	from fully discharged, 45 °C, until 4.2 V	4.63 Ah = 81.9 % SOC in ~836 s (charge-side capacity)	[PASS] PASS (design target 4.6 Ah)

Items explicitly N/A (beyond pure simulation boundary / outside task scope, requires physical experiment):
nail penetration; crush; drop; cycle life (aging model available on Chen2020 but not part of this task); -20 °C low-temperature discharge; 5C rate discharge.

Conclusion

All task criteria pass for C7: energy density 457.93 Wh/kg $\geq$ 392.61; 4C charge at 45 °C with T_max 318.79 K $\leq$ 333.15 K and no lithium plating (SPMe + DFN cross-check); overcharge to 4.7 V without thermal runaway. Uncovered items listed above — the plating margin under DFN is +5 mV, so a physical-prototype 4C validation is recommended (paper limitation, stated honestly).